

Vasudev Mittal

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Research Interests

Testing the Cosmological Principle; Large Scale Structure of the Universe; Cosmic Dipole Tension; Bayesian Statistics; Data-driven Cosmology

Education

Doctor of Philosophy

Sydney Institute for Astronomy, School of Physics

Thesis title: *Confronting the Cosmological Principle*

Supervisor: Prof. Geraint F. Lewis

University of Sydney

October 2024 -

Bachelor of Science - Masters of Science (BS-MS)

Integrated Master's degree program

Physics majors and Astronomy minor

(Top 3 in the graduating batch)

IISER Mohali

August 2019 - June 2024

Publications

- Mittal, Vasudev; Lewis, Geraint F.; **The Ellis and Baldwin test of the Cosmic Dipole: Exploring the impact of multiple flux density cuts**, Journal of Cosmology and Astroparticle Physics, DOI:
- Mittal, Vasudev; Oyada, Oliver T.; Lewis, Geraint F.; **Cosmic multipoles in galaxy surveys – II. Comparing different methods in assessing the cosmic dipole**, Open Journal of Astrophysics, DOI: 10.33232/001c.144907
- Oyada, Oliver T.; Mittal, Vasudev; Lewis, Geraint F.; **Cosmic multipoles in galaxy surveys – I. How inferences depend on source counts and masks**, Monthly Notices of the Royal Astronomical Society, DOI: 10.1093/mnras/stae2776
- Oyada, Oliver T.; Mittal, Vasudev; Lewis, Geraint F.; Murphy, Tara; **A Bayesian approach to the cosmic dipole in radio galaxies: joint analysis of NVSS & RACS**, Monthly Notices of the Royal Astronomical Society, DOI: 10.1093/mnras/stae1399
- Mittal, Vasudev; Oyada, Oliver T.; Lewis, Geraint F.; **The Cosmic Dipole in the Quia Sample of Quasars: A Bayesian Analysis**, Monthly Notices of the Royal Astronomical Society, DOI: 10.1093/mnras/stad3706
- Mittal, Vasudev; **Low Scientific Research Output in India: A Critical Analysis of NEP, 2020 Regarding Scientific Research**, Research Journal of Social and Life Sciences, Vol-32, Year-16, June 2020, Impact Factor: 3.93, ISSN: 0973-3914

In Preparation

- Mittal, Vasudev; et. al. (alphabetical author list) The Cosmic Radio Dipole and Observer Effects in SKA all-sky surveys; SKA Science Book.

Conference Presentations

- **Confronting the Cosmological Principle** in The Non-Gaussian Universe: Advanced Statistical Inference in the Era of Next-Generation Cosmological Surveys held on June 16-19, 2026.
- **Is the Universe Isotropic?-Maybe!!** (*Runner-up by popular vote*) in CosmoVerseSchool@Sofia held on May 25-29, 2026.
- **Confronting the Cosmological Principle: Shaking the foundation of the cosmos** (*Awarded First Prize in Oral Presentation*) in 44th International Workshop on Bayesian Inference and Maximum Entropy Methods in Science and Engineering held on December 14-19, 2025.
- **Polarimetric Analysis of Hermite-A Lunar Crater using Chandrayaan-2 DFSAR Data** in 3rd Indian Planetary Science Conference held on March 14-16, 2022.
- **Equation of State of Dark Energy from Generalized Trace Dynamics** (*Awarded First Prize in Oral Presentation*) in 8th National Students Symposium on Physics held on Nov. 12-14, 2021.

Notable Talks

Seminar Universe - Institut d'Astrophysique de Paris (IAP)	June 2026
SAC - Indian Space Research Organization (ISRO) [<i>Invited talk</i>]	January 2026
I. Institute of Physics, University of Cologne	July 2025
SPASE Department, IIT Kanpur	April 2025

Teaching

Mechanics Lab <i>First Year Undergraduate Course</i>	<i>Tutor for Main Stream</i> <i>February - June 2026</i>
Quantum mechanics and electromagnetic properties of matter <i>Second Year Undergraduate Course</i>	<i>Tutor for Main Stream</i> <i>August-November 2025</i>
Optics and thermodynamics <i>Second Year Undergraduate Course</i>	<i>Tutor for Advanced Stream</i> <i>February - June 2025</i>

Peer review services

Monthly Notices of the Royal Astronomical Society	Since May 2025
Scientific Reports, a <i>Nature</i> Portfolio Journal	Since April 2025

Visiting Research Positions

- **Visiting Research Student** at [The University of Sydney, Australia](#). (10 Aug - 07 Nov, 2023)
- **DAAD-WISE Fellow** at [Bielefeld University, Germany](#). (15 May-31 July, 2023)
- **Visiting Student** at [Inter University Center for Astronomy and Astrophysics](#). (1 Jun -10 Jul, 2022)
- **INSPIRE-SHE Fellow** at [Tata Institute of Fundamental Research, India](#) (May-Aug 2022)
- **Science Academy Summer Research Fellow** at [Shiv Nadar University](#). (1 Jul-25 Aug, 2021)

- **ISRO-TREES Fellow** in the [Indian Space Research Organization, India](#). (13 May-12 Sep, 2021)

Significant Achievements

- **Physics Foundation Scholarship 2024**, fully funded PhD scholarship awarded by the Physics Foundation, University of Sydney.
- **Lindau Nobel Laureates Meeting 2024**, selected as a young scientist participant for the meeting.
- **All India Rank 49 in CSIR-NET (Physics, 34000 applicants)**, the National eligibility test for Assistant Professorship, also eligible for becoming a Junior Research Fellow with CSIR.
- **DAAD-WISE Fellowship 2023 (≈ 150 selections out of ≈ 5000 applicants)**, awarded by the German academic exchange service for undertaking summer research in Germany
- Listed among the **top 1 % (all India)** in Class XII (Sr. Sec. High School) CBSE Board Examination 2019 with a percentage of 97%.

Schools and Workshops Attended

- **CosmoVerse School @ Sofia**, organised by the CosmoVerse COST initiative (May 2026).
- **73 Lindau Nobel Laureate Meeting (Physics)**, organized by the Lindau Council (July 2024).
- **17 Kosmologietag**, organized by the ZiF-Bielefeld University, Germany. (June 2023).
- **Less Traveled Path to the Dark Universe**, organized by the ICTS-TIFR, India. (March 2023).
- **Asian Science Camp 2022**, organized by the ASC association and IBS, South Korea (July 2022).
- **International Summer School on the ISM**, organized by the CNRS, France (July 2021).
- **National Science Camp 2019**, organized by DST-INSPIRE at IISER Kolkata (December 2019).

Memberships of Academic Bodies

Life Member	Indian Association for General Relativity and Gravitation
Life Member	The Indian Association of Physics Teachers
Member	Astronomical Society of India
Member	Astronomical Society of Australia